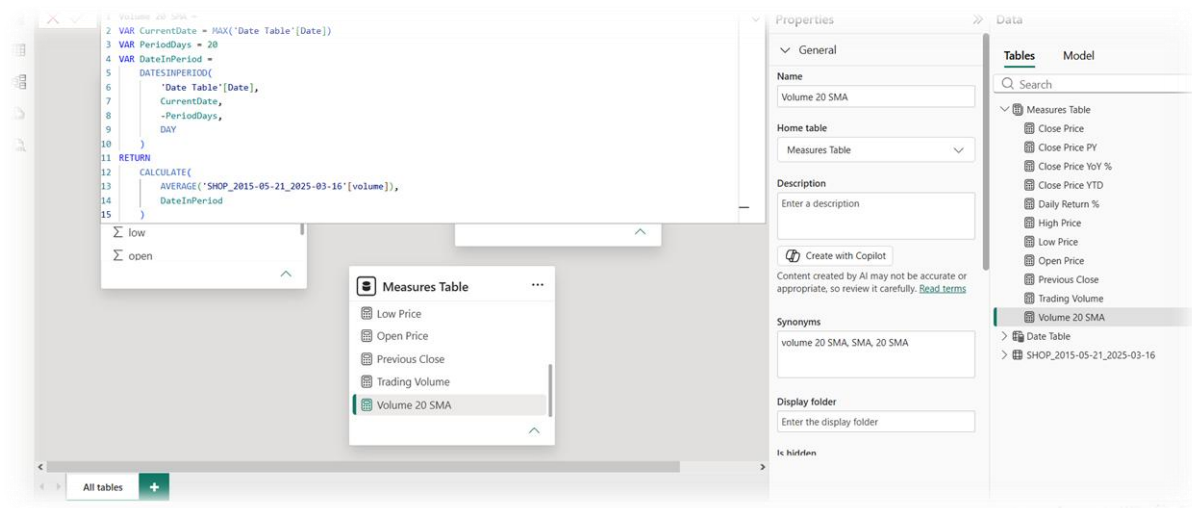


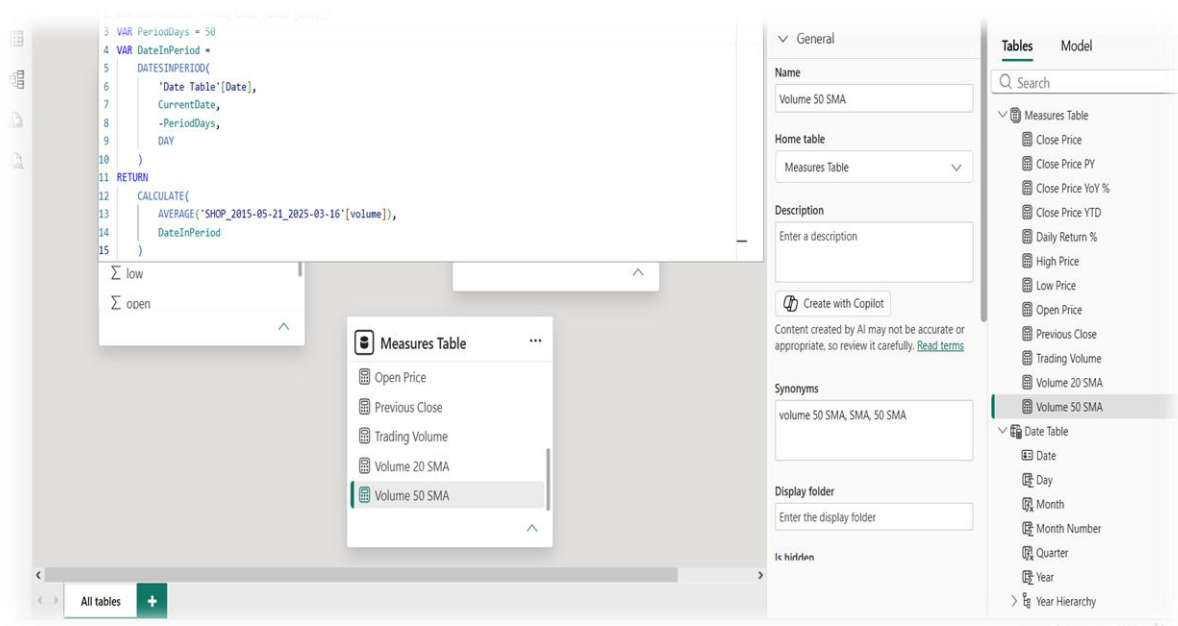
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Volume 20-Day Simple Moving Average

A **20-Day Simple Moving Average (Volume 20 SMA)** was implemented using DAX to calculate the average trading volume over the preceding 20-day period. The **DATESINPERIOD()** function was used to define the required time window, allowing short-term changes in trading activity to be analysed.

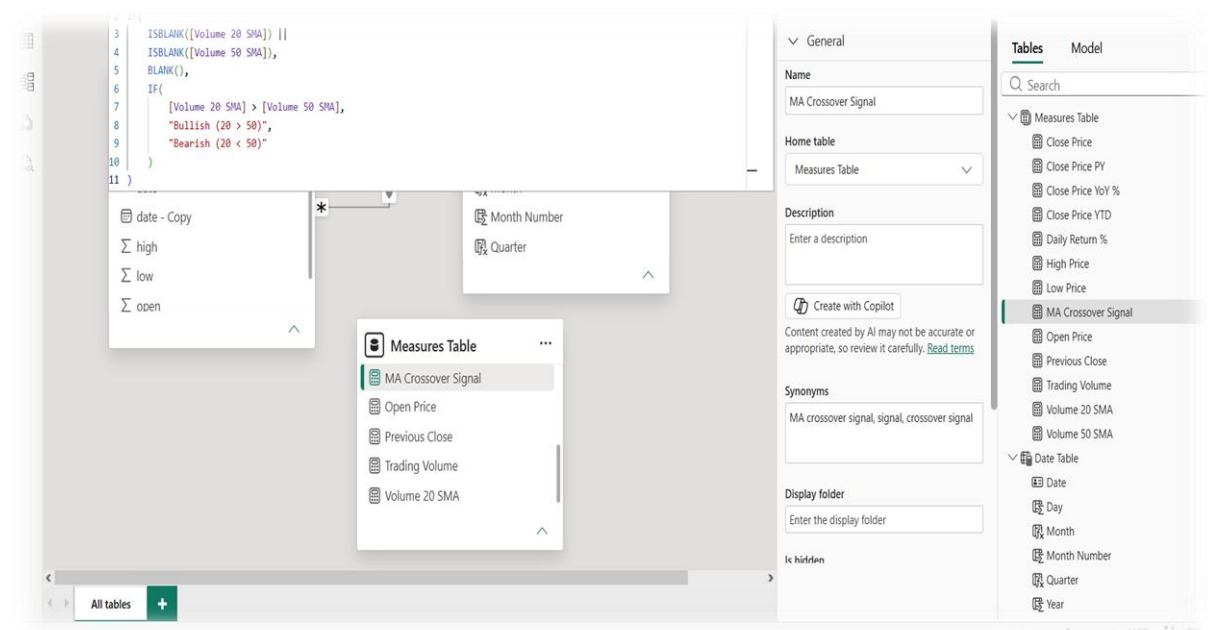


The **Volume 20 SMA** measure was created using DAX. The **DATESINPERIOD()** function was used to calculate the average trading volume for the previous 20 days. This helps in comparing the current trading volume with the recent average.



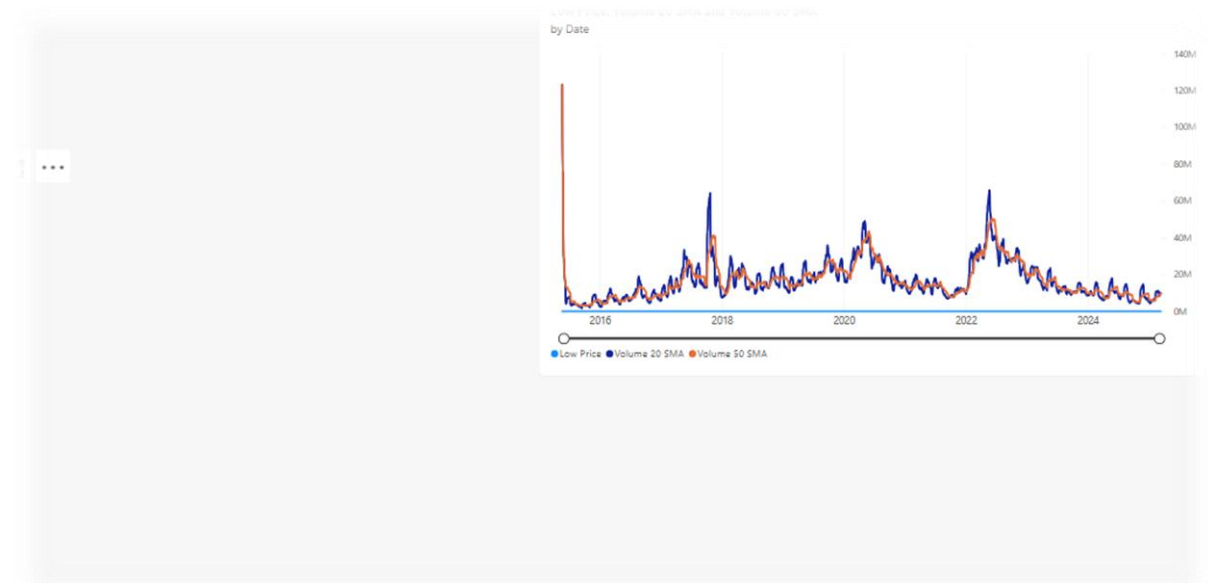
Volume 50 SMA

The **Volume 50 SMA** was created using the same method, but for a period of 50 days. It gives a longer-term view of the trading volume and helps in understanding the overall volume trend.



MA Crossover Signal

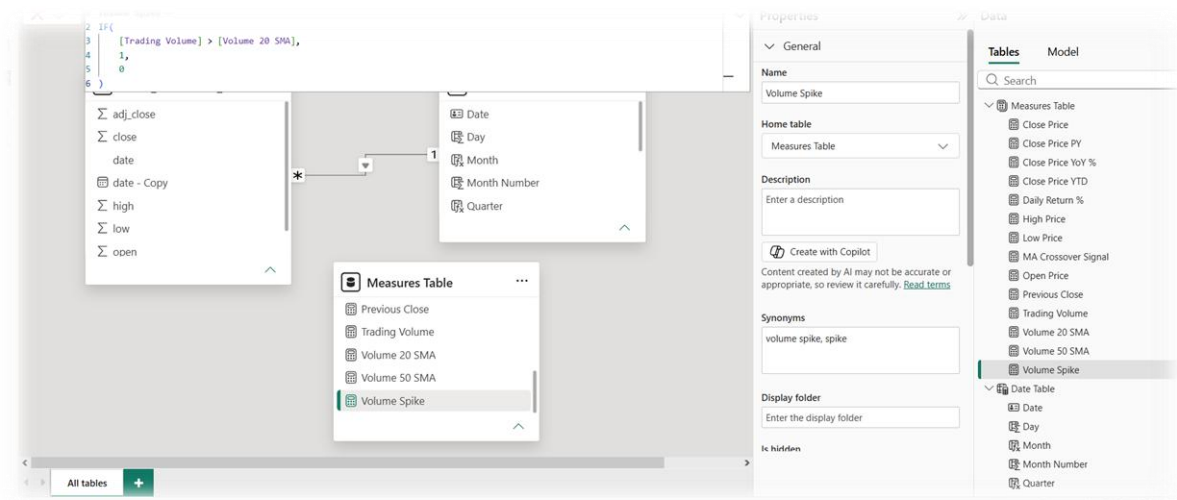
The **MA Crossover Signal** measure was created to compare the 20-day SMA with the 50-day SMA. It shows "**Bullish (20 > 50)**" when the 20-day SMA is higher than the 50-day SMA. Otherwise, it shows "**Bearish (20 < 50)**".



Stock Price and Moving Average Chart

A line chart was created using the **Date** field on the X-axis. The chart contains **Close Price**, **Volume 20 SMA** and **Volume 50 SMA**. This helps to compare the stock price with the two moving averages over time.

The chart settings and filters were also checked to make sure that all the required measures were included correctly.



A line chart was created using the **Date** field on the X-axis. The chart contains **Close Price**, **Volume 20 SMA** and **Volume 50 SMA**. This helps to compare the stock price with the two moving averages over time.

The chart settings and filters were also checked to make sure that all the required measures were included correctly.

Zoom Slider

The **Zoom Slider** option was enabled from the formatting settings of the chart. It allows the user to zoom into a particular part of the stock-price timeline for a closer view.

Trading Volume Chart

Another chart was created to show **Trading Volume** and **Volume 20 SMA** over time. This makes it easier to compare the actual trading volume with its 20-day average.

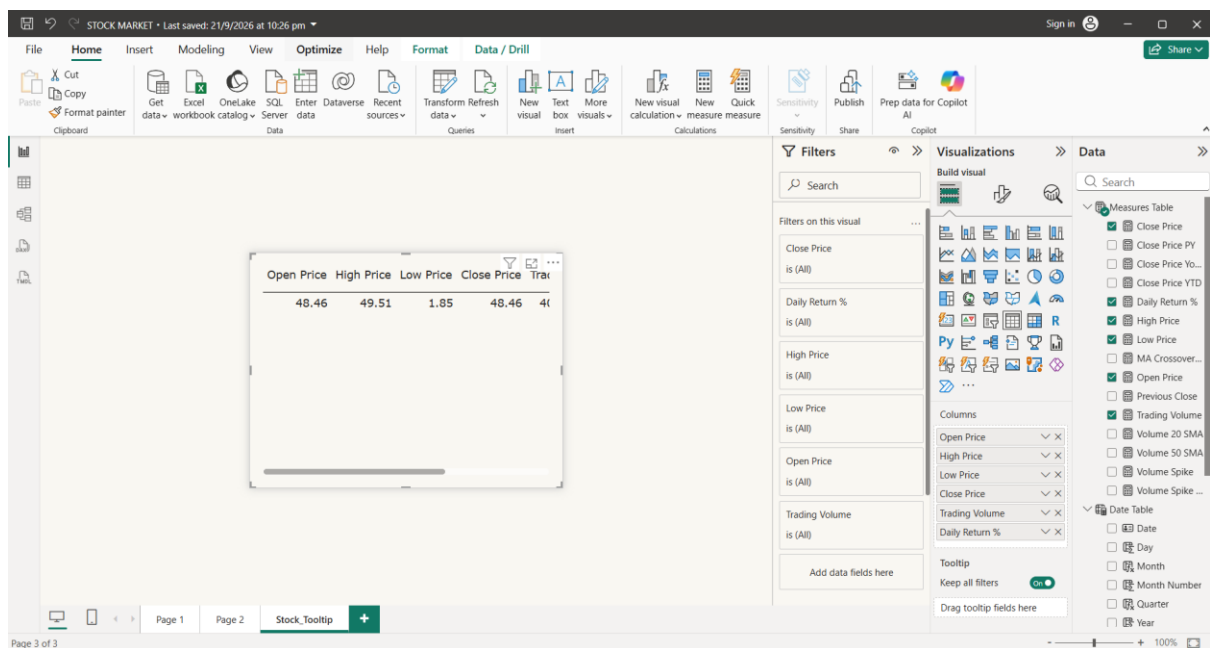
Volume Spike

A **Volume Spike** DAX measure was created to identify the dates where the **Trading Volume** is greater than the **Volume 20 SMA**. This helps to find periods with unusually high trading activity.

Volume Spike Highlight

The **Volume Spike Highlight** measure was created to show the trading volume only when it is higher than the 20-day SMA. This makes the volume spikes easier to identify in the chart.

The measure was then added to the volume chart along with **Trading Volume** and **Volume 20 SMA** so that the high-volume periods could be seen clearly.



Date Range Slicer

A **Date Range Slicer** was added using the Date Table[Date] field. It was set to **Between** mode, allowing the user to select a start date and an end date.

This slicer controls the time period shown in the stock-price and trading-volume charts.

Relative Date Slicer

A **Relative Date Slicer** was also added using the Date Table[Date] field. It was set to the **Relative Date** option, which allows the user to filter the data based on a selected relative time period.

This is useful for analysing recent stock data without having to manually select the start and end dates.

Stock Price Metric

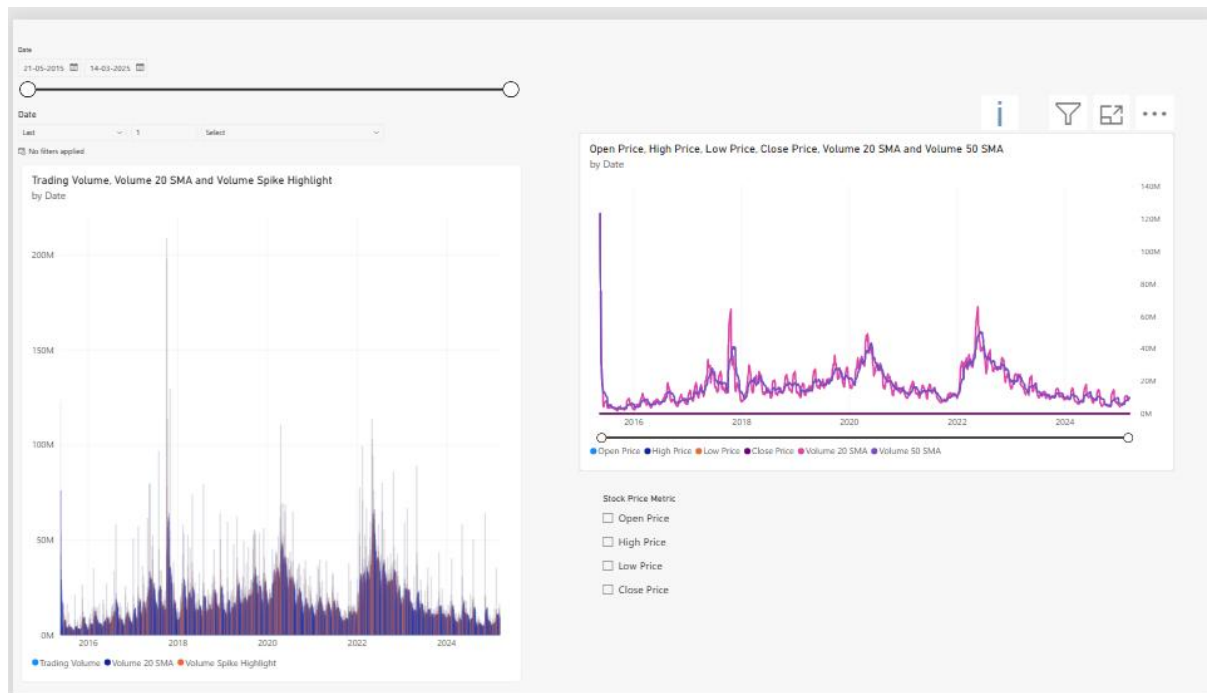
A **Field Parameter** named **Stock Price Metric** was created using **Open Price**, **High Price**, **Low Price** and **Close Price**. This allows the user to select which stock-price measure should be displayed in the chart.

The generated parameter was added to the model with all four price measures as the available options.

Dynamic Stock Price Chart

The **Stock Price Metric** parameter was added to the main chart. The user can now switch between **Open Price**, **High Price**, **Low Price** and **Close Price** using the parameter.

The **20-day SMA** and **50-day SMA** remain in the chart, so the selected stock-price measure can be compared with the moving averages.



Final Dashboard

The final dashboard combines the **stock-price chart**, **moving averages**, **trading-volume chart**, **volume spike highlighting**, **date slicers** and **Stock Price Metric selector**.

These features make the report interactive and allow the user to analyse stock prices and trading volume for different time periods.